

Math 3108 - Section 103 (Murphy) - Fall 2019 - Exam 3

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Instructions.

- No outside notes or resources.
 - Silence your phones completely (including vibrations).
 - Be mindful of your classmates.
 - No calculators. You need to show your work.
 - There are 6 problems for a total of 100 points.
 - You have 50 minutes to complete the exam.
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1. [20 points]

(i) Let

$$A = \begin{bmatrix} -5 & 4 \\ -8 & 7 \end{bmatrix}.$$

Find the eigenvalues of A . For each eigenvalue, find a corresponding eigenvector.

$$A - \lambda I = \begin{bmatrix} -5 & 4 \\ -8 & 7 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} -5-\lambda & 4 \\ -8 & 7-\lambda \end{bmatrix}$$

$$\det(A - \lambda I) = (-5-\lambda)(7-\lambda) + 32 = 0$$

$$-35 + 5\lambda - 7\lambda + \lambda^2 + 32 = 0$$

$$\lambda^2 - 2\lambda - 3 = 0$$

$$(\lambda - 3)(\lambda + 1) = 0$$

$$\boxed{\lambda = 3, -1}$$

$$A - 3I = \begin{bmatrix} -5 & 4 \\ -8 & 7 \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} -8 & 4 \\ -8 & 4 \end{bmatrix} \sim \begin{bmatrix} -8 & 4 \\ 0 & 0 \end{bmatrix} \sim \begin{bmatrix} -2 & 1 \\ 0 & 0 \end{bmatrix}$$

$$-2x_1 + x_2 \rightarrow 2x_1 = x_2 \quad \boxed{\begin{bmatrix} 1 \\ 2 \end{bmatrix} \text{ for } \lambda = 3}$$

$$A + I = \begin{bmatrix} -5 & 4 \\ -8 & 7 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -4 & 4 \\ -8 & 8 \end{bmatrix} \sim \begin{bmatrix} -4 & 4 \\ 0 & 0 \end{bmatrix} \sim \begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix}$$

$$-x_1 + x_2 \rightarrow x_1 = x_2 \quad \boxed{\begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ for } \lambda = -1}$$

(ii) Let

$$A = \begin{bmatrix} 0 & 8 \\ -2 & 0 \end{bmatrix}.$$

Find the eigenvalues of A . For each eigenvalue, find a corresponding eigenvector.

$$A - \lambda I = \begin{bmatrix} 0 - \lambda & 8 \\ -2 & 0 - \lambda \end{bmatrix} \quad \det(A - \lambda I) = (-\lambda)(-\lambda) + 16 = \lambda^2 + 16$$

$$\frac{-0 \pm \sqrt{0^2 - 4(1)(16)}}{2(1)} = \frac{\pm \sqrt{-64}}{2} = \frac{\pm 8i}{2} = \boxed{\pm 4i = \lambda}$$

$$A - (4i)I = \begin{bmatrix} 0 & 8 \\ -2 & 0 \end{bmatrix} - \begin{bmatrix} 4i & 0 \\ 0 & 4i \end{bmatrix} = \begin{bmatrix} -4i & 8 \\ -2 & -4i \end{bmatrix} \quad \begin{aligned} (-4i)x_1 + 8x_2 &= 0 \\ (4i)x_1 &= 8x_2 \\ (i)x_1 &= 4x_2 \end{aligned}$$

by complex conjugate $\rightarrow \boxed{\begin{bmatrix} -i \\ 4 \end{bmatrix}} \text{ for } \lambda = -4i$

$\rightarrow \boxed{\begin{bmatrix} i \\ 4 \end{bmatrix}} \text{ for } \lambda = 4i$

2. [20 points] Consider the following matrix:

$$A = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & 0 \\ -1 & -1 & 3 \end{bmatrix}.$$

The characteristic polynomial of A is $p(\lambda) = (\lambda - 1)(\lambda - 2)^2$. $\rightarrow \lambda = 1, 2$ (mult: 2)
Determine whether or not A is diagonalizable. Explain your answer.

$$\begin{aligned} A - 2I &= \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & 0 \\ -1 & -1 & 3 \end{bmatrix} - \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} -1 & -1 & 1 \\ 0 & -1 & 0 \\ -1 & -1 & 1 \end{bmatrix} \sim \begin{bmatrix} -1 & -1 & 1 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ &\sim \begin{bmatrix} -1 & 0 & 1 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

Not diagonalizable - for A to be diagonalizable, there must be an eigenvector for each eigenvalue and their multiplicities. However, there are not two free variables, so $\dim \text{Nul}(A - 2I) \neq$ the multiplicity of $\lambda = 2$.

3. [20 points] Let

$$w_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 2 \end{bmatrix}, \quad w_2 = \begin{bmatrix} 5 \\ 10 \\ -5 \\ 0 \end{bmatrix}, \quad \text{and} \quad A = [w_1 \ w_2] = \begin{bmatrix} 1 & 5 \\ 0 & 10 \\ 0 & -5 \\ 2 & 0 \end{bmatrix}.$$

(i) Use the Gram-Schmidt algorithm to find an orthogonal basis for $\text{Span}\{w_1, w_2\}$.

$$v_1 = w_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 2 \end{bmatrix}$$

$$v_2 = w_2 - \frac{w_2 \cdot v_1}{v_1 \cdot v_1} v_1 = \begin{bmatrix} 5 \\ 10 \\ -5 \\ 0 \end{bmatrix} - \frac{\begin{bmatrix} 5 \\ 10 \\ -5 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \\ 2 \end{bmatrix}}{\begin{bmatrix} 1 \\ 0 \\ 0 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \\ 2 \end{bmatrix}} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 2 \end{bmatrix}$$

Orthogonal basis:

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 4 \\ 10 \\ -5 \\ -2 \end{bmatrix} \right\}$$

$$= \begin{bmatrix} 5 \\ 10 \\ -5 \\ 0 \end{bmatrix} - \frac{1}{5} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \\ -5 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 10 \\ -5 \\ -2 \end{bmatrix}$$

(ii) Find a basis for $[\text{Col}(A)]^\perp$.

$$[\text{Col}(A)]^\perp = \text{Nul}(A^T)$$

$$A^T = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 5 & 10 & -5 & 0 \end{bmatrix} \xrightarrow{-5R_1 + R_2 \rightarrow R_2} \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 10 & -5 & -10 \end{bmatrix} \xrightarrow{\frac{1}{10}R_2 \rightarrow R_2}$$

$$\begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & -\frac{1}{2} & -1 \end{bmatrix}$$

$$x_1 + 2x_4 = 0$$

$$x_2 - \frac{1}{2}x_3 - x_4 = 0$$

$$\rightarrow x_1 = -2x_4$$

$$x_2 = \frac{1}{2}x_3 + x_4$$

x_3 free

x_4 free

$$\vec{x} = \begin{bmatrix} -2x_4 \\ \frac{1}{2}x_3 + x_4 \\ x_3 \text{ free} \\ x_4 \text{ free} \end{bmatrix}$$

$$\left\{ \begin{bmatrix} 0 \\ \frac{1}{2} \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ 0 \\ 1 \end{bmatrix} \right\} = \mathcal{B}_{[\text{Col}(A)]^\perp}$$

4. [20 points] The system

$$Ax = b, \text{ where } A = \begin{matrix} & v_1 & v_2 \\ \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ 1 & 1 \end{bmatrix} & \text{and } b = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix} \end{matrix}$$

is inconsistent. Find the least squares solution(s) to $Ax = b$.

Hint. The columns of A are orthogonal.

$$A\hat{x} = \hat{b}$$

$$\hat{b} = \frac{b \cdot v_1}{v_1 \cdot v_1} v_1 + \frac{b \cdot v_2}{v_2 \cdot v_2} v_2$$

$$= \frac{\begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}}{\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} + \frac{\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}}{\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$= \frac{3 - 1 + 1}{1 + 1 + 1} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} + \frac{0 + 1 + 1}{0 + 1 + 1} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} = \hat{b}$$

$$A\hat{x} = \hat{b}$$

$$\left[\begin{array}{cc|c} 1 & 0 & 1 \\ -1 & 1 & 0 \\ 1 & 1 & 2 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & 0 & 1 \\ -1 & 1 & 0 \\ 0 & 2 & 2 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 2 & 2 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{array} \right]$$

$$\boxed{\hat{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}}$$

5. [10 points]

(i) TRUE or FALSE

(-6)

If A and B are similar matrices, then A and B have the same eigenvectors.

(ii) TRUE or FALSE

If an $n \times n$ matrix A has n distinct real eigenvalues, then A is diagonalizable.

(iii) TRUE or FALSE

If S is an orthogonal set of vectors, then S is linearly independent.

6. [10 points]

(i) Give an example of a 2×2 matrix that is invertible but not diagonalizable.

$$\boxed{\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}} \quad (1-\lambda)(1-\lambda) = 0 \quad \lambda = 1 \quad A - I = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

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only one free variable
(need two)

(ii) Suppose A is a 2×2 matrix with the following eigenvalue/eigenvector pairs:

$$\lambda_1 = 1, \quad \underline{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \lambda_2 = 2, \quad \underline{v}_2 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

Find a diagonal matrix D and an invertible matrix P such that $A = PDP^{-1}$.

$$\boxed{P = \begin{bmatrix} 1 & 3 \\ 1 & 1 \end{bmatrix}} \rightarrow P^{-1} = \frac{1}{1-3} \begin{bmatrix} 1 & -3 \\ -1 & 1 \end{bmatrix} = -\frac{1}{2} \begin{bmatrix} 1 & -3 \\ -1 & 1 \end{bmatrix}$$

$$\boxed{D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}}$$

$$A = PDP^{-1} = -\frac{1}{2} \begin{bmatrix} 1 & 3 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & -3 \\ -1 & 1 \end{bmatrix}$$